

Randomized Kolmogorov–Smirnov Calibration for Fitted Discrete Single-Cell Residual Diagnostics

Elvis Han Cui^{a,b,*}, Yihao Li^{a,c}, Zhuang Liu^{a,d}

^a*College of Environmental and Resource Sciences, Zhejiang University, Hangzhou, China*

^b*Department of Biostatistics, University of California, Los Angeles, CA, USA*

^c*Genentech, South San Francisco, CA, USA*

^d*Google, Mountain View, CA, USA*

Abstract

Residual diagnostics for discrete single-cell models often reuse the data used to fit a count margin, trend, or simulator. *Calibration* means choosing a reference law that preserves the nominal false-positive rate for this fitted procedure. The ordinary Kolmogorov–Smirnov (KS) reference generally fails because it treats the null distribution as fixed. We specify a randomized residual workflow that separates marginal, covariate-indexed, and dependence diagnostics and repeats simulation, fitting, and residual construction in the bootstrap. Classical composite-null empirical-process theory supplies the justification. Known discrete margins yield independent uniform residuals for independent observations; regular fitted margins yield a plug-in Gaussian limit, targeted by the bootstrap under corresponding consistency conditions. Separate gene-wise marginal KS tests cannot detect copula-only errors. Simulations through 10,000 observations show persistent miscalibration of naive references and near-nominal rejection after refitting. scGTM and scDesign3 examples illustrate distinct residual targets, while public IMvigor210 PBMC data expose discrepancies from a patient-and-depth conditional working model. The contribution is a discrete-residual specialization of established theory with explicit limits for complex fitted models.

Keywords: Kolmogorov–Smirnov statistic, single-cell simulation, randomized probability integral transform, empirical process, parametric bootstrap, copula diagnostics

2020 MSC: 62G10, 62G20, 60F17, 62P10

*Corresponding author.

Email addresses: elviscuihan@g.ucla.edu (Elvis Han Cui), li.yihao@gene.com (Yihao Li), zhuangl@google.com (Zhuang Liu)

1. Introduction

Single-cell count models are routinely fitted to observed cells and then checked using residuals computed from those same cells. This occurs in interpretable trajectory models such as scGTM and in the fitted margins and dependence components used by the scDesign family (Li and Li, 2019; Sun et al., 2021; Cui et al., 2022; Song et al., 2024). A randomized probability integral transform (PIT) is attractive in this setting: under a known discrete count distribution it converts each count to a uniform residual, after which a Kolmogorov–Smirnov (KS) statistic has its familiar fixed-null reference law. The same conclusion does not follow when the count distribution has first been estimated from the observations being checked.

The same distinction arises in translational biomarker studies. Genentech investigators used single-cell RNA sequencing of baseline peripheral blood mononuclear cells (PBMCs) from 10 patients in the phase 2 IMvigor210 atezolizumab program to study IL-8-associated myeloid inflammation and antigen presentation (Rosenberg et al., 2016; Yuen et al., 2020). The per-patient integer count matrices are publicly available under GEO GSE145281. They provide a clinically sourced setting in which patient identity, sequencing depth, and cellular heterogeneity are part of the fitted-reference problem, even when the diagnostic target is only a conditional gene-wise count margin.

This distinction creates a practical failure mode. Suppose a negative-binomial margin is fitted gene by gene, randomized PIT residuals are computed, and the ordinary uniform KS cutoff is applied. The fitted residuals reuse the data to estimate their own reference distribution and are therefore not an independent sample from a fixed uniform law. In the simulation in Section 5, this procedure has a rejection rate near zero at a nominal level of 5%, whereas a bootstrap that repeats the fitting step restores rejection close to the target. Overly large critical values can also conceal model discrepancies. The problem is that fitting changes the distribution of the statistic while the fixed-null reference ignores that change.

Throughout the paper, *calibration* means selecting the reference law, critical value, or Monte Carlo p -value for the complete procedure that produced the statistic. The procedure includes estimation, residual randomization, aggregation across genes or covariate strata, and any simulator refitting. Calibration is distinct from model adequacy: a calibrated test can still have

low power, and a uniform marginal residual distribution does not establish correct gene–gene dependence or conditional structure.

Calibration is also distinct from numerical optimization. Nature-inspired methods have been used to maximize the scGTM likelihood, including the competitive swarm optimizer with mutated agents (CSO-MA) (Cui et al., 2024). Such methods address how the count model is fitted; residual calibration addresses how departures from that fitted model are assessed. Even an accurately optimized likelihood does not make the resulting fitted residuals an independent uniform sample. This distinction connects the scGTM modeling and optimization literature to the diagnostic problem considered here.

The ingredients needed to analyze this problem are largely classical. Randomized distributional transforms and quantile residuals handle discrete margins; composite-null KS theory accounts for parameter estimation; parametric bootstrap methods approximate fitted-null reference laws; and Rosenblatt or copula diagnostics distinguish marginal from joint fit (Rosenblatt, 1952; Dunn and Smyth, 1996; Lilliefors, 1967; Durbin, 1973; Khmaladze, 1981; Babu and Rao, 2004). Existing composite-null theory provides the necessary framework. Applying it still requires specifying the residual transform, the estimation steps included in the reference procedure, and the model component that the statistic can assess.

This paper makes four contributions. First, it specifies an operational workflow that maps a scientific diagnostic target to a residual statistic and the corresponding fixed-null or fitted-null reference law. Second, it specializes the classical composite-null empirical-process expansion to randomized PIT residuals for fitted discrete margins and states conditions under which a full refitted bootstrap targets that process. Third, it formalizes the limitation of separately interpreted gene-wise marginal KS statistics for copula-only misspecification. Fourth, simulations, scGTM and scDesign3 examples, and a public IMvig210 baseline-PBMC analysis show how marginal, covariate-indexed, and dependence diagnostics lead to different conclusions. These are a discrete-residual specialization and an application-level synthesis; we do not claim a new general composite-null empirical-process theorem or a new single-cell simulator.

Table 1 distinguishes the classical foundations from this paper’s contributions. We consider models with an explicit discrete or conditional distribution from which randomized PIT residuals can be computed. High-dimensional latent-variable and foundation models require separate justification of these conditions.

Table 1: Classical ingredients and the specific additions made in this paper.

Component	Established foundation	Addition and role in this paper
Discrete residuals	Randomized PIT and randomized quantile residuals	Explicit fitted-versus-known reference distinction for discrete single-cell margins.
Composite-null KS	Durbin–Babu–Rao empirical-process expansion	Specialization to the randomized residual map and a component-specific interpretation.
Parametric bootstrap	Simulation from an estimated null model	An operational requirement to repeat the complete fit, randomization, and statistic.
Joint diagnostics	Rosenblatt and copula diagnostics	An exact statement of what separate gene-wise marginal KS tests cannot establish.
Single-cell application	scGTM and scDesign fitted count models	A unified workflow separating trajectory, marginal, covariate-indexed, and dependence conclusions.

2. Diagnostic Target and Operational Workflow

The workflow begins by specifying the model component to be assessed. Let $D = \{(X_i, Z_i) : 1 \leq i \leq n\}$ contain count vectors X_i and observed covariates Z_i , and let $\mathcal{A}$ denote the complete fitting algorithm. The fitted object may be a gene-wise margin, a pseudotime-dependent count model, or a simulator containing both margins and a dependence model. For a fitted conditional distribution $F_{g,\hat{\theta}}(\cdot | Z_i)$, the randomized residual for gene g is

$$U_{ig} = F_{g,\hat{\theta}}(X_{ig} - | Z_i) + V_{ig} p_{g,\hat{\theta}}(X_{ig} | Z_i), \quad V_{ig} \sim \text{Unif}(0, 1).$$

The randomizers are generated after fitting and are independent of the data.

Figure 1 displays the fitted-model procedure. The workflow returns a component-specific diagnostic conclusion rather than a single claim that the complete simulator is valid.

The distinction between a fixed-parameter and a refitted bootstrap is essential. Simulating from $F_{\hat{\theta}}$ but retaining $\hat{\theta}$ when the bootstrap residuals are computed targets a different procedure. Likewise, fitting gene-wise margins in the observed data but using an oracle margin in the bootstrap omits the estimation effect that requires calibration.

Algorithm 1 (Randomized residual calibration for a fitted workflow). Given data D , fitting algorithm $\mathcal{A}$, residual transform T , scalar diagnostic statistic S , number of bootstrap samples B , and significance level α :

1. Fit the observed data using the complete algorithm, $\hat{\theta} = \mathcal{A}(D)$.
2. Generate independent PIT randomizers, construct the observed randomized residuals $U = T(D, \hat{\theta}, V)$, and compute $S_{\text{obs}} = S(U, D)$.
3. If every distribution entering T is fully specified under the null, and the sampling and independence conditions for S hold, use the corresponding fixed-null reference law and stop. A reference estimated from a separate training sample is not automatically a known null distribution.
4. Otherwise, for $b = 1, \dots, B$:
 - (a) simulate D_b^* from the fitted model at $\hat{\theta}$ while preserving the conditioning design used by the analysis;
 - (b) refit with the same algorithm, $\hat{\theta}_b^* = \mathcal{A}(D_b^*)$;
 - (c) generate fresh independent randomizers V_b^* and compute $S_b^* = S\{T(D_b^*, \hat{\theta}_b^*, V_b^*), D_b^*\}$.
5. Report

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbf{1}(S_b^* \geq S_{\text{obs}})}{B + 1}$$

or the empirical $(1 - \alpha)$ quantile of $S_1^*, \dots, S_B^*$. State the diagnostic target and do not extend the conclusion to untested components.

For a global diagnostic over many genes, repeat the same scalar aggregation in S_{obs} and every S_b^* . For separate gene-wise tests, first calibrate each statistic and then apply the prespecified multiplicity rule.

The plus-one rule avoids reporting zero Monte Carlo p -values; its resolution is $1/(B + 1)$. It does not make a fitted parametric bootstrap finite-sample exact. Its inferential justification depends on the model, sampling design, and bootstrap conditions stated in Section 3. A small p -value flags a discrepancy in the tested component; a large value does not establish that the component, or the full simulator, is correct.

3. Theoretical Justification

3.1. Fixed-null baseline

The classical finite-sample KS reference law applies to independent uniforms under a fully specified null. On the uniform scale, the one-sample

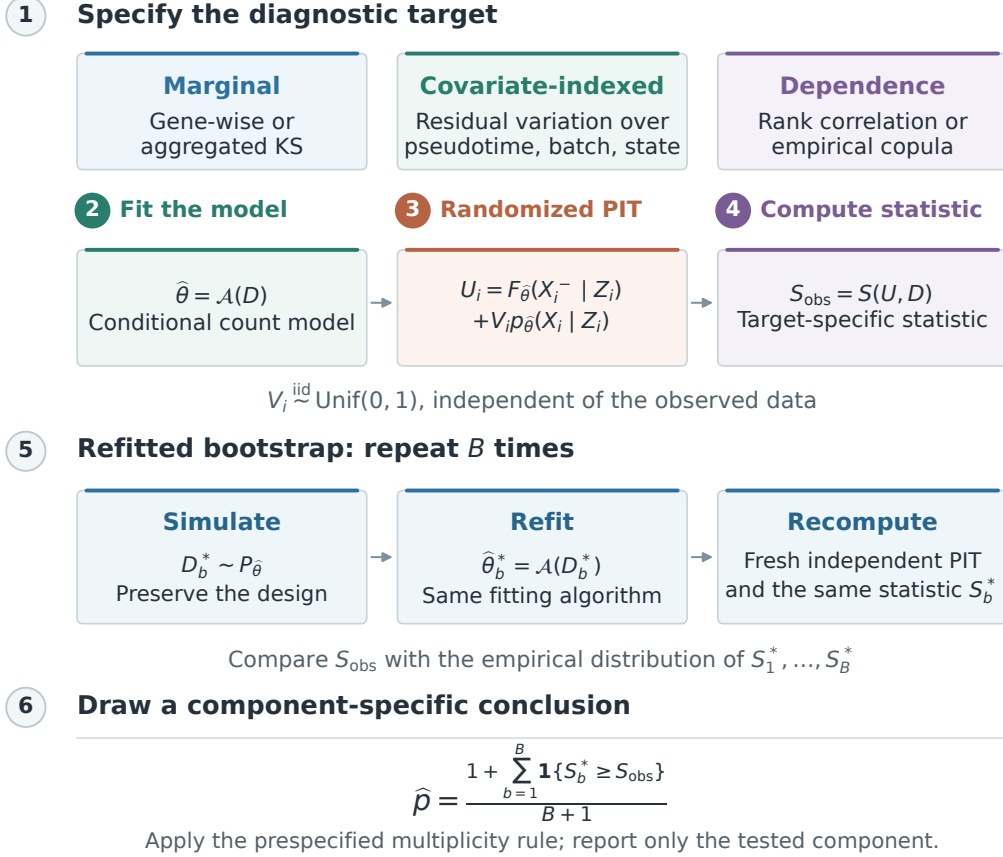


Figure 1: Randomized-PIT diagnostic workflow for a fitted model. Data D contain counts X and covariates Z ; $\mathcal{A}$ is the fitting algorithm. The chosen target determines the statistic: separate marginal KS statistics do not test dependence. Bootstrap calibration repeats the same fitting and diagnostic operations, including fresh residual randomization. If the null is fully specified, the appropriate fixed-null law replaces steps 2 and 5, provided its sampling conditions hold. Fitted-model bootstrap validity requires the regularity and consistency conditions stated in Section 3.

empirical path drifts between observations and jumps at observations; exact one-sided and two-sample probabilities can therefore be written as finite crossing or path-counting calculations. These fixed-null formulas are the baseline against which fitted-simulator diagnostics should be judged. The detailed crossing formulas, recursions, and limiting fixed-null identities are given in Supplementary Appendix A–D.

The point needed for single-cell simulators is simple. If a known count

Table 2: Diagnostic targets, reference laws, and the conclusions supported by the workflow.

Target	Example statistic	Reference	Supported conclusion	Not established
Marginal count fit	Gene-wise or aggregated uniform KS	Fixed law for known margins; refitted bootstrap for fitted margins	Evidence of discrepancies in the specified marginal count distributions	Gene-gene dependence or covariate structure
Conditional fit	Residual process indexed by pseudotime, batch, or cell state	Refitted bootstrap preserving the conditioning design	Evidence of residual variation over the indexed covariate	Unindexed dependence components
Dependence fit	Residual rank correlation, empirical copula, or Rosenblatt statistic	Joint fitted-model simulation with refitting	Evidence of discrepancies in the tested dependence feature	All higher-order dependence unless included in the statistic

margin is applied to independent draws and transformed by the randomized probability integral transform, the transformed residuals are independent uniforms and the ordinary KS crossing law applies. If the same margin is estimated first, the residual path is no longer the fixed-null empirical bridge. The fitting step must be included in the reference law, either analytically through a fitted empirical-process limit or computationally through a refitted bootstrap. We state this fitted-null principle first for ordinary composite models and then specialize it to randomized residuals for scDesign-style count simulators.

3.2. Composite nulls and parameter estimation

Everything in the fixed-null argument relies on knowing the null distribution before seeing the data. Once parameters are estimated from the same data, the transformed observations no longer behave like independent uniforms under a fixed distribution. The path being tested has been altered by the fitting step, so the ordinary distribution-free calibration no longer applies.

Let $X_1, \dots, X_n$ be independent observations from F_θ , with unknown parameter vector θ . For a normal distribution, $\theta = (\mu, \sigma^2)^T$. Estimate θ by maximum likelihood or another regular method, and define

$$Z_n(t, \theta) = \sqrt{n} (F_n(t) - F_\theta(t)) \quad (1)$$

$$Z_n(t, \hat{\theta}) = \sqrt{n} (F_n(t) - F_{\hat{\theta}}(t)). \quad (2)$$

The key step in classical composite-null KS theory is a first-order expansion of the fitted distribution (Durbin, 1973; Babu and Rao, 2004; Van der Vaart,

2000; Del Barrio et al., 2007):

$$\begin{aligned}
Z_n(t, \hat{\theta}) &= Z_n(t, \theta) + \sqrt{n} (F_\theta(t) - F_{\hat{\theta}}(t)) \\
&= Z_n(t, \theta) - \frac{\partial F_\theta(t)^T}{\partial \theta} \sqrt{n}(\hat{\theta} - \theta) + o_{\mathbb{P}_\theta}(1) \\
&= Z_n(t, \theta) - \frac{\partial F_\theta(t)^T}{\partial \theta} \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_\theta(X_i) + o_{\mathbb{P}_\theta}(1)
\end{aligned}$$

Here the estimator is assumed to admit the expansion

$$\sqrt{n}(\hat{\theta} - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_\theta(X_i) + o_{\mathbb{P}_\theta}(1),$$

The influence function ψ_θ describes each observation's first-order contribution to the estimator and has mean zero and finite second moment. The fitted-process limit follows from the joint limit of

$$\left(Z_n(\cdot, \theta), \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_\theta(X_i) \right).$$

Donsker's theorem gives $Z_n(\cdot, \theta) \Rightarrow B(F_\theta(\cdot)) = B \circ F_\theta$, where B is a standard Brownian bridge, while the central limit theorem gives $\frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_\theta(X_i) \Rightarrow \mathcal{N}(0, \Sigma(\theta))$, where $\Sigma(\theta) = \mathbb{E}\psi_\theta(X_1)^{\otimes 2}$. The covariance can be computed from

$$\text{Cov}(\mathbb{I}(X_1 \leq t), \psi_\theta(X_1)) = \int_{-\infty}^t \psi_\theta(x) F_\theta(dx) := h(t, \theta).$$

Hence, we have finite-dimensional weak convergence. Tightness then yields weak convergence of the stochastic process $Z_n(t, \hat{\theta})$ (Babu and Rao, 2004); the limit is a mean-zero Gaussian process $Z(\cdot, \theta)$ with covariance

$$\begin{aligned}
\text{Cov}(Z(s, \theta), Z(t, \theta)) &= F_\theta(s \wedge t) - F_\theta(s)F_\theta(t) \\
&\quad - h(t, \theta)^T \frac{\partial F_\theta(s)}{\partial \theta} \\
&\quad - h(s, \theta)^T \frac{\partial F_\theta(t)}{\partial \theta} \\
&\quad + \frac{\partial F_\theta(s)^T}{\partial \theta} \Sigma(\theta) \frac{\partial F_\theta(t)}{\partial \theta}.
\end{aligned} \tag{3}$$

The following theorem states the conditions needed for this expansion.

Theorem 3.1 (Durbin-Babu-Rao ([Durbin, 1973](#); [Babu and Rao, 2004](#))). *Let $X_1, \dots, X_n$ be a random sample with common distribution function F_θ and $\theta \in \mathbb{R}^k$. Let $\hat{\theta}$ be an estimator of θ with influence function ψ_θ . Assume that*

$$\sqrt{n}(\hat{\theta} - \theta) = n^{-1/2} \sum_{i=1}^n \psi_\theta(X_i) + o_{\mathbb{P}_\theta}(1),$$

that $F_\theta(t)$ is uniformly differentiable in θ along $\hat{\theta} - \theta$, so that

$$\sup_t \left| \sqrt{n}\{F_{\hat{\theta}}(t) - F_\theta(t)\} - \frac{\partial F_\theta(t)}{\partial \theta}^T \sqrt{n}(\hat{\theta} - \theta) \right| = o_{\mathbb{P}_\theta}(1),$$

and that, with P_n denoting the empirical measure and P_θ the law under F_θ , the empirical process and estimator expansion converge jointly,

$$\left(\sqrt{n}(P_n - P_\theta)\mathbf{1}\{X \leq \cdot\}, n^{-1/2} \sum_{i=1}^n \psi_\theta(X_i) \right) \Rightarrow (B_\theta, Z_\theta)$$

in $\ell^\infty(\mathcal{T}) \times \mathbb{R}^k$, where $\mathcal{T}$ is the index set over which the KS supremum is taken. Define

$$\Sigma(\theta) = \mathbb{E}_\theta\{\psi_\theta(X)\psi_\theta(X)^T\}, \quad h(t, \theta) = \mathbb{E}_\theta[\{\mathbf{1}(X \leq t) - F_\theta(t)\}\psi_\theta(X)].$$

Assume the joint limit has cross-covariance $\text{Cov}\{B_\theta(t), Z_\theta\} = h(t, \theta)$, and define

$$D_n(\hat{\theta}) = \sup_t |F_n(t) - F_{\hat{\theta}}(t)|.$$

Then, at every continuity point ϵ of the limiting law,

$$\mathbb{P}\left(\sqrt{n}D_n(\hat{\theta}) > \epsilon\right) \longrightarrow \mathbb{P}\left(\sup_t |Z(t, \theta)| > \epsilon\right),$$

where

$$Z(t, \theta) = B_\theta(t) - \frac{\partial F_\theta(t)}{\partial \theta}^T Z_\theta$$

is a mean-zero Gaussian process with covariance specified in (3). Bootstrap or Monte Carlo approximations require the corresponding consistency conditions for this fitted-null process.

Proof. This is the classical composite-null expansion; a derivation in the notation used here is given in Supplementary Appendix F.

Remark 3.2 (Consequences of naive KS calibration). If one estimates the parameter and then uses the classical Brownian-bridge critical value as if θ were known, the test uses an inappropriate reference law. Table 3 illustrates the resulting conservatism at nominal level 5% with $n = 100$. The classical cutoff is $1.358/\sqrt{n}$; the bootstrap column uses a parametric bootstrap in which parameters are re-estimated in every bootstrap sample. Implementation details are given in Supplementary Appendix H.

Table 3: Empirical rejection probabilities under the null, based on 2000 Monte Carlo samples. The bootstrap column uses 399 bootstrap resamples per Monte Carlo sample.

Null model	KS cutoff	Refit bootstrap
Known $N(0, 1)$	0.051	—
Fitted normal (μ, σ^2)	0.000	0.044
Fitted exponential rate	0.007	0.052

Remark 3.3 (Cases outside the linear expansion). There are also models where the Durbin-Babu-Rao expansion itself should not be used without extra work. A simple example is a distribution with parameter-dependent support, such as $\text{Unif}(0, \theta)$ with θ estimated by the sample maximum: the estimator has an n -rate, non-Gaussian limit and is not asymptotically linear in the form assumed above. Mixture models near nonidentifiable parameter values create a different version of the same problem. In such cases, a refitted simulation scheme may still be useful, but the limiting Gaussian projection in (3) is no longer the right justification.

Remark 3.4. In the case of maximum likelihood estimation, the influence function is

$$\psi_\theta(x) = I_\theta^{-1} s_\theta(x),$$

where I_θ^{-1} is the inverse of Fisher information and $s_\theta(x)$ is the score function evaluated at x .

Example. Let $X \sim N(\mu, \sigma^2)$. The influence function for the maximum likelihood estimator is

$$\psi_\theta(x) = \begin{pmatrix} x - \mu \\ (x - \mu)^2 - \sigma^2 \end{pmatrix},$$

and

$$\Sigma(\theta) = I_\theta^{-1} = \begin{pmatrix} \sigma^2 & 0 \\ 0 & 2\sigma^4 \end{pmatrix}.$$

3.3. Randomized residual processes for fitted discrete models

Single-cell trend models and count simulators add one more layer to the fitted-null problem. Counts are discrete, so the ordinary probability integral transform is not uniform without randomization. Moreover, trend and simulator parameters are fitted before the diagnostic is computed. This section gives the resulting calibration rules for fitted count margins, including scGTM-style conditional trends and scDesign-style margins and copulas. The conditions below have three practical roles: the estimator must have a stable first-order approximation, the residual distribution must vary smoothly with the fitted parameter, and the empirical fluctuation must remain controlled when that parameter is substituted. A Donsker condition is one way to obtain a Gaussian approximation uniformly over residual thresholds. These properties must be checked for the chosen model and fitting procedure.

Theorem 3.5 (Residual empirical-process calibration). *Let (X_i, W_i) be independent copies of (X, W) , where W is an auxiliary randomizer independent of X . The estimator is computed from the observed X_i 's before the auxiliary randomizers are generated. Let $\{T_\theta : \theta \in \Theta\}$ be measurable maps from the data-randomizer space into $(0, 1)$, and suppose that $T_{\theta_0}(X, W) \sim \text{Unif}(0, 1)$. Write P_n for the empirical measure of (X_i, W_i) and P_{θ_0} for the law of (X, W) at the true parameter. Let $\hat{\theta}$ satisfy*

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{\theta_0}(X_i) + o_{\mathbb{P}}(1).$$

Define

$$H_\theta(u) = \mathbb{P}_{\theta_0}\{T_\theta(X, W) \leq u\},$$

$$\alpha_n(u) = \sqrt{n} \left\{ \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{T_{\hat{\theta}}(X_i, W_i) \leq u\} - u \right\}.$$

Assume:

1. the class $\{\mathbf{1}(T_{\theta_0} \leq u) : 0 < u < 1\}$ is Donsker;
2. the local plug-in empirical-process remainder satisfies

$$\sup_{0 < u < 1} \sqrt{n} |(P_n - P_{\theta_0})\{\mathbf{1}(T_{\hat{\theta}} \leq u) - \mathbf{1}(T_{\theta_0} \leq u)\}| = o_{\mathbb{P}}(1);$$

3. the deterministic centering map is locally uniformly linear along $\widehat{\theta}$, namely

$$\sup_{0 < u < 1} \left| \sqrt{n} \{H_{\widehat{\theta}}(u) - H_{\theta_0}(u)\} - \dot{H}_{\theta_0}(u)^T \sqrt{n}(\widehat{\theta} - \theta_0) \right| = o_{\mathbb{P}}(1);$$

4. the process $\sqrt{n}(P_n - P_{\theta_0})\mathbf{1}\{T_{\theta_0} \leq u\}$ in $\ell^\infty(0, 1)$ and $\sqrt{n}(\widehat{\theta} - \theta_0)$ converge jointly to (B, Z_θ) .

Then in $\ell^\infty(0, 1)$,

$$\alpha_n(u) \Rightarrow B(u) + \dot{H}_{\theta_0}(u)^T Z_\theta,$$

where B is a Brownian bridge and Z_θ is the Gaussian limit of $\sqrt{n}(\widehat{\theta} - \theta_0)$, jointly distributed with B . Consequently, if

$$K_n = \sup_{0 < u < 1} |\alpha_n(u)|,$$

then at continuity points c of the law of $\sup_{0 < u < 1} |B(u) + \dot{H}_{\theta_0}(u)^T Z_\theta|$,

$$\mathbb{P}(K_n \leq c) \rightarrow \mathbb{P} \left\{ \sup_{0 < u < 1} |B(u) + \dot{H}_{\theta_0}(u)^T Z_\theta| \leq c \right\}.$$

Proof. For the random $\widehat{\theta}$, $H_{\widehat{\theta}}(u)$ means the deterministic map $\theta \mapsto H_\theta(u)$ evaluated at $\widehat{\theta}$. Add and subtract the process at the true parameter:

$$\alpha_n(u) = \sqrt{n}(P_n - P_{\theta_0})\mathbf{1}\{T_{\theta_0} \leq u\} + R_n(u) + \sqrt{n}\{H_{\widehat{\theta}}(u) - u\},$$

where

$$R_n(u) = \sqrt{n}(P_n - P_{\theta_0})\{\mathbf{1}(T_{\widehat{\theta}} \leq u) - \mathbf{1}(T_{\theta_0} \leq u)\}.$$

The first term converges to a Brownian bridge because $T_{\theta_0}(X, W)$ is uniform and the indicator class is Donsker. Assumption 2 gives $\sup_u |R_n(u)| = o_{\mathbb{P}}(1)$. Since $H_{\theta_0}(u) = u$, Assumption 3 gives the uniformly valid expansion

$$\sqrt{n}\{H_{\widehat{\theta}}(u) - u\} = \dot{H}_{\theta_0}(u)^T \sqrt{n}(\widehat{\theta} - \theta_0) + o_{\mathbb{P}}(1)$$

in $\ell^\infty(0, 1)$. The joint convergence in Assumption 4 completes the proof. $\square$

This theorem applies the classical expansion to a specified residual map; it does not verify regularity for a particular simulator. The map T_θ may be a randomized PIT for a count simulator, a Rosenblatt-type transform for a

copula, or a residual transform for a fitted conditional count model, once the listed empirical-process conditions have been verified for that transform. The local plug-in remainder in Assumption 2 is the substantive condition: it is where one verifies that estimating θ does not add an unaccounted indicator-class fluctuation beyond the displayed derivative term. In smooth finite-dimensional settings with fixed support this is typically proved from stochastic equicontinuity of the threshold class and root- n consistency of the estimator. For penalized, black-box, nonregular, clustered, or high-dimensional simulator fits, this condition is not automatic. Repeating the fitting algorithm in a bootstrap does not establish it or prove bootstrap consistency. When the required conditions have not been verified, the refitted reference is an empirical approximation whose accuracy needs a model-specific assessment. The ordinary KS law is recovered in the fixed-parameter case, and in fitted cases only under additional degeneracy or cancellation, for example a vanishing first-order derivative term.

For an integer-valued distribution function F , write $F(k-) = \mathbb{P}(X < k)$ and $p_F(k) = F(k) - F(k-)$. For $v \in (0, 1)$ define the randomized distributional transform

$$T_F(k, v) = F(k-) + v p_F(k).$$

This is the form of the randomized transform used for discrete margins in the scDesign2 copula construction, up to the equivalent convention for the uniform randomizer (Sun et al., 2021, 2023).

Proposition 3.6 (Exact marginal crossings for known discrete margins). *Let $X_1, \dots, X_n$ be independent observations from a known integer-valued count distribution F on a countable ordered support, and let $V_1, \dots, V_n$ be independent $\text{Unif}(0, 1)$ variables independent of the counts. Set $U_i = T_F(X_i, V_i)$. The resulting variables are independent and each has the $\text{Unif}(0, 1)$ law. Consequently their KS statistics have the ordinary simple-null uniform-scale calibration; in particular, the one-sided statistic has the fixed-null crossing law summarized in Section 3.1 and detailed in the supplementary appendix.*

Proof. Conditional on $X_i = k$, U_i is uniform on the interval $(F(k-), F(k)]$. The intervals with $p_F(k) > 0$ partition $(0, 1)$ up to endpoints and have lengths $p_F(k)$. For any $u \in (0, 1)$, choose a count k with $p_F(k) > 0$ and $F(k-) \leq u \leq F(k)$; at shared endpoints either adjacent positive-mass interval

gives the same value. Then

$$\begin{aligned}\mathbb{P}(U_i \leq u) &= \sum_{\ell < k} p_F(\ell) + p_F(k) \mathbb{P}\{F(k-) + V_i p_F(k) \leq u \mid X_i = k\} \\ &= F(k-) + u - F(k-) = u.\end{aligned}$$

Thus each U_i is uniform on $(0, 1)$. Independence follows because the pairs (X_i, V_i) are independent. The exact KS crossing formulas then apply to the transformed sample. $\square$

The preceding proposition is exact but it assumes that the margin is known. Fitted single-cell simulators do not have this property. The next result is the discrete residual analogue of the fitted-null empirical-process calculation in Section 3.2.

Corollary 3.7 (Plug-in randomized bridge under regular fitted margins). *Let $\{F_\theta : \theta \in \Theta \subset \mathbb{R}^q\}$ be a parametric family of discrete count distributions and suppose the data are generated from F_{θ_0} . Let $\hat{\theta}$ be an estimator satisfying*

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{\theta_0}(X_i) + o_{\mathbb{P}}(1).$$

For independent $V_i \sim \text{Unif}(0, 1)$, define $U_i(\theta) = T_{F_\theta}(X_i, V_i)$ and

$$\alpha_n(u) = \sqrt{n} \left\{ \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{U_i(\hat{\theta}) \leq u\} - u \right\}, \quad 0 < u < 1.$$

Let

$$G_\theta(u) = \mathbb{P}_{\theta_0}\{T_{F_\theta}(X, V) \leq u\}$$

and assume that the four conditions in Theorem 3.5 hold for this transform, with $H_\theta = G_\theta$. In particular, this assumes that the local plug-in empirical-process remainder is negligible and that the deterministic centering map is locally uniformly linear along $\hat{\theta}$, with derivative $\dot{G}_{\theta_0}(u)$. Then in $\ell^\infty(0, 1)$,

$$\alpha_n(u) \Rightarrow B(u) + \dot{G}_{\theta_0}(u)^T Z_\theta,$$

where B is a Brownian bridge and Z_θ is the Gaussian limit of $\sqrt{n}(\hat{\theta} - \theta_0)$, jointly distributed with B .

Proof. Substitute the randomized discrete transform into Theorem 3.5; details are given in Supplementary Appendix F.

Corollary 3.8 (Refitted simulator bootstrap under conditional expansion). *Assume the conditions of Corollary 3.7 and, conditional on the observed data, the bootstrap analogues of the Donsker/tightness condition, local plug-in remainder, local centering linearity, asymptotic linearity, and joint convergence conditions. More explicitly, assume also that $\hat{\theta} \rightarrow_{\mathbb{P}} \theta_0$, that the conditional joint law of the bootstrap empirical process and $\sqrt{n}(\hat{\theta}^* - \hat{\theta})$ converges in probability to the same Gaussian pair (B, Z_{θ}) as in Corollary 3.7, and that the derivative $\dot{G}_{\hat{\theta}}^*(u)$ of the conditional bootstrap centering map satisfies*

$$\sup_{0 < u < 1} \|\dot{G}_{\hat{\theta}}^*(u) - \dot{G}_{\theta_0}(u)\| \rightarrow_{\mathbb{P}} 0.$$

Conditional on the observed data $\mathcal{D}_n$, generate

$$X_i^* \stackrel{\text{ind}}{\sim} F_{\hat{\theta}}, \quad V_i^* \stackrel{\text{ind}}{\sim} \text{Unif}(0, 1), \quad i = 1, \dots, n,$$

with the bootstrap counts and uniforms conditionally independent. Refit the parameter to obtain $\hat{\theta}^$ and compute*

$$K_n^* = \sup_{0 < u < 1} \left| \sqrt{n} \left\{ \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{T_{F_{\hat{\theta}^*}}(X_i^*, V_i^*) \leq u\} - u \right\} \right|.$$

Then the conditional distribution of K_n^ consistently estimates the limiting distribution of $K_n = \sup_{0 < u < 1} |\alpha_n(u)|$: for every continuity point c of the limiting law,*

$$\mathbb{P}^*(K_n^* \leq c \mid \mathcal{D}_n) - \mathbb{P}(K_n \leq c) \rightarrow_{\mathbb{P}} 0.$$

If the bootstrap sample is generated from $F_{\hat{\theta}}$ but the parameter is held fixed when the statistic is recomputed, the bootstrap calibrates the process obtained by holding that fitted margin fixed, rather than the complete fitted workflow. It omits the first-order estimation term except in degenerate cases where that term vanishes.

Proof. Apply the residual-process expansion conditionally under the bootstrap law. The full conditional expansion is given in Supplementary Appendix F.

Proposition 3.9 (A marginal KS blind spot for copula simulators). *Consider independent cells from a `scDesign2`-style simulator with fixed margins*

$F_1, \dots, F_p$ and copula parameter R . Let R' be another copula parameter with the same margins. For any fixed gene j , the randomized residuals

$$U_{ij} = T_{F_j}(X_{ij}, V_{ij}), \quad i = 1, \dots, n,$$

have the same marginal distribution under R and R' . Hence any diagnostic or rejection rule measurable only with respect to the empirical distribution of $U_{1j}, \dots, U_{nj}$ has the same rejection probability under R and R' ; it has no power beyond its size against alternatives that change only the dependence structure. This statement does not cover procedures that use the joint law of multiple gene-wise residual processes across paired cells.

Proof. A copula changes the joint distribution within a cell while preserving all univariate margins. Proposition 3.6 converts the j th margin to uniform residuals regardless of the copula parameter, and the cells are independent by assumption. This is only a statement about a fixed gene’s marginal residual process; it does not assert cross-gene independence. Therefore a single-gene KS process has the same distribution under R and R' . Across cells, independence makes the empirical distribution a sample from the same univariate uniform law under both copulas. This proves only a single-gene marginal statement, not a claim about diagnostics that combine paired residuals across genes. $\square$

Remark 3.10. Proposition 3.9 does not say that copula misspecification is invisible. It says that it is invisible to a single-gene marginal KS diagnostic. Dependence diagnostics must use joint residual structure, for example pair-wise residual ranks, empirical copula processes, or two-dimensional crossing statistics on (U_j, U_k) .

Remark 3.11 (Conditional residual diagnostics). Let Z_i denote a cell-level covariate such as cell type, pseudotime, batch, modality, or spatial location. This includes pseudotime trend models such as scGTM and conditional simulator margins such as those fitted by scDesign3 (Cui et al., 2022; Song et al., 2024). Condition on the observed covariates, or assume independent cells with covariates Z_i . If a known conditional count margin $F_{\theta_0}(\cdot \mid Z_i)$ is correctly specified, then

$$U_i = T_{F_{\theta_0}(\cdot \mid Z_i)}(X_i, V_i)$$

satisfies $U_i \mid Z_i \sim \text{Unif}(0, 1)$, and the residuals are conditionally independent when the cells are. Thus residual non-uniformity within covariate regions, or

residual association between U_i and Z_i , is a population lack-of-fit signal; after fitting, it is descriptive unless calibrated by the fitted conditional workflow. The statement treats Z_i as fixed or external to the count-margin fit. If pseudotime, cell type, batch labels, or other covariates are estimated from the same expression data, that estimation step becomes part of the fitted workflow and must be included in the empirical-process expansion or in the refitted bootstrap.

With fitted conditional parameters, the analogous residual process is a plug-in empirical process indexed by both the residual threshold and the covariate region. A formal version requires indexed empirical-process conditions of the same type as above, such as a manageable covariate class, local centering linearity of the conditional margin, and a joint limit for the fitted conditional parameter. Accordingly, simple uniform KS summaries are descriptive unless they are calibrated by the fitted conditional workflow, for example by a refitted bootstrap.

4. Interpretation for Fitted Simulator Diagnostics

The fitted-null limit can be read geometrically for efficient estimation in regular parametric models: fitting removes empirical-process variation in score directions, so the residual bridge is a plug-in or projected Gaussian process rather than the ordinary Brownian bridge. This interpretation is useful, but it is not needed to run the diagnostic. In practice, the operational requirement is to calibrate the exact statistic generated by the fitted simulator workflow, including the same estimation, randomization, and residual transformation steps used on the observed data. Supplementary Appendix E gives the geometric and finite crossing-complex interpretation.

5. Simulation Studies and Real-Data Diagnostics

5.1. Simulation Studies

We evaluate the effect of parameter estimation on randomized-PIT KS tests using independent negative-binomial counts with mean 6, size parameter $r = 2.5$, and variance $6 + 6^2/r$. At $n = 250, 1,000$, and $10,000$, we compare four reference procedures: the ordinary KS cutoff with known parameters, the same cutoff after moment fitting, a fixed-parameter bootstrap, and a refitted bootstrap. The nominal level is 5%. Power is assessed by adding a 15% point mass at zero to the generating distribution. Separate experiments examine

conditional-mean and copula misspecification. Supplementary Appendix H gives the estimation rules and implementation details.

Table 4: Negative-binomial randomized residual calibration across sample sizes. The null Monte Carlo sample sizes are 400, 250, and 300, and the corresponding bootstrap sizes are 199, 149, and 99. Power is evaluated against a negative-binomial distribution mixed with a 15% point mass at zero using 200, 125, and 100 Monte Carlo samples. “Fixed boot” simulates from the fitted model but does not refit it; “refit boot” repeats parameter estimation. The four type-I error columns are empirical null rejection proportions at nominal level 0.05. Power is the refitted-bootstrap rejection proportion under the zero-inflated alternative.

n	Empirical type-I error					Runtime (s)
	Known	Naive KS	Fixed boot	Refit boot	Refit power	
250	0.052	0.000	0.000	0.062	0.955	13.5
1,000	0.056	0.000	0.000	0.040	1.000	12.9
10,000	0.047	0.000	0.003	0.040	1.000	68.4

Across the three sample sizes, the ordinary KS cutoff after fitting yields no rejections, and the fixed-parameter bootstrap yields rejection rates of 0.0–0.3% (Table 4). Refitting yields rejection rates of 4.0–6.2%, close to the nominal 5%, with power of 95.5–100% against the zero-inflated alternative. These results show that parameter re-estimation, not simulation alone, is needed to reproduce the null variation of the fitted residual statistic in this setting.

The estimation effect need not disappear from the reference law as n increases. Although $\hat{\theta} - \theta_0 = O_{\mathbb{P}}(n^{-1/2})$, the KS process is also multiplied by $\sqrt{n}$. Its estimation term therefore remains $O_{\mathbb{P}}(1)$:

$$\alpha_n(u) = B_n(u) + \dot{G}_{\theta_0}(u)^T \sqrt{n}(\hat{\theta} - \theta_0) + o_{\mathbb{P}}(1),$$

where B_n denotes the empirical process at the true margin. The unscaled discrepancy shrinks, but the difference between fixed-null and fitted-null critical values can persist on the scale used for testing. The rejection rates in Table 4 illustrate this distinction; their Monte Carlo uncertainty is reported in Supplementary Appendix H.

The copula block implements Proposition 3.9. Samples are generated from two negative-binomial margins joined by a Gaussian copula with data-generating correlation $\rho = 0.65$. When the synthetic sample uses the wrong copula correlation but the same margins, including the moderate $\rho = 0.50$ change and the stronger $\rho = 0$ independence misspecification, the two gene-wise marginal KS diagnostics remain near the nominal level. A residual

Table 5: Component-specific simulations. Panel A uses two independent 240-cell samples per trial, known NB margins, and 1000 trials; brackets are 95% Wilson intervals. Panel B uses scDesign3 NB regressions, 500 outer samples per scenario, and 199 bootstrap refits per sample; brackets are exact 95% binomial intervals. The conditional statistic is the maximum of all six scaled between-quartile KS distances. “Fixed” omits refitting; “refit” repeats the complete fit and statistic. The two diagnostic targets are reported separately, not combined into one rejection rule. The Periodic / linear row reports power; the other Panel B rows report size.

A. Copula misspecification with known margins				
Scenario		Gene 1 KS	Gene 2 KS	Correlation
Same ($\rho = 0.65$)		0.043 [0.032;0.057]	0.052 [0.040;0.068]	0.058 [0.045;0.074]
Wrong ($\rho = 0.50$)		0.054 [0.042;0.070]	0.050 [0.038;0.065]	0.663 [0.633;0.692]
Independent ($\rho = 0$)		0.039 [0.029;0.053]	0.049 [0.037;0.064]	1.000 [0.996;1.000]

B. Fitted conditional margins: generating / fitted mean				
Scenario	n	Marginal refit	Conditional fixed	Conditional refit
Linear / linear	240	0.064 [0.044;0.089]	0.022 [0.011;0.039]	0.052 [0.034;0.075]
Periodic / linear	240	0.084 [0.061;0.112]	0.874 [0.842;0.902]	0.924 [0.897;0.946]
Periodic / periodic	240	0.048 [0.031;0.071]	0.006 [0.001;0.017]	0.064 [0.044;0.089]
Spline / spline	2087	0.060 [0.041;0.085]	0.002 [0.000;0.011]	0.064 [0.044;0.089]

correlation diagnostic, however, reacts to dependence misspecification, with partial power for the moderate change and near-certain power for the independence case. The correlation reference is approximate; this known-margin experiment is not a test of fitted copula calibration.

Panel B evaluates the complete conditional-margin procedure using scDesign3’s NB regression function. On fixed pseudotimes, we simulate log-linear, periodic, and cubic-spline mean curves and re-estimate both the mean and NB size in every bootstrap sample, including smoothing for the spline fit. The two statistics are the marginal KS and the maximum of six scaled between-quartile KS distances. Each maximum is calibrated as one statistic; six unadjusted pairwise tests are not used. The sample size is 240 for the linear and periodic models and 2,087 for the spline null, whose fitting configuration matches the pancreatic conditional margins. Supplementary Appendix H.5 specifies the generating curves, fitting rules, and independent observed and bootstrap streams. Across the three correctly specified cases, refitted marginal rejection rates range from 4.8% to 6.4%, and conditional rejection rates from 5.2% to 6.4%. When the periodic component is omitted, conditional rejection is 92.4%, compared with 8.4% for the marginal statistic. The two diagnostic targets thus have different sensitivity to the omitted trend in this setting. These are finite-sample checks of the specified regressions, with Monte Carlo

uncertainty shown in the table, not a proof of validity for arbitrary fitted simulators. All 398,000 bootstrap refits and 2,000 observed fits succeeded.

Table 6: scGTM residual diagnostics for representative genes and the full 10-gene demonstration. The fitted change time t_0 and trend strengths k_1, k_2 describe the trajectory; D is the uniform KS distance and Bin KS is the largest two-sample distance between pseudotime quartiles. Simple p -values are descriptive because the margins were fitted. All genes use the hill-shaped trend without the package’s valley transformation. Supplementary Appendix H identifies the complete gene-level results.

Gene	Margin	t_0	k_1	k_2	ϕ	D	Simple p	Bin KS
Gene5	Poisson	0.425	4.995	3.554	—	0.357	1.28×10^{-57}	0.352
	NB	0.454	4.593	4.802	1.980	0.025	0.905	0.128
Gene10	Poisson	0.643	2.790	5.282	—	0.244	1.03×10^{-26}	0.312
	NB	0.616	2.958	4.658	3.052	0.041	0.373	0.200
10-gene aggregate		median D : Poisson 0.278, NB 0.035; NB lower in 10/10 genes; median Bin KS decrease 0.112						

scGTM trend-model demonstration. We use the count matrix distributed in the scGTM software demonstration, comprising 500 observations with supplied pseudotimes, and fit its first 10 genes. This software example is not treated as independent biological validation; its generating mechanism is not documented in the source repository. Supplementary Appendix H identifies the input and software version. Figures 2 and 3 and Table 6 connect the residual calibration principle to an interpretable pseudotime trend model. We retain the scGTM package’s particle-swarm fitting algorithm for this example. The fit reports a change time t_0 and side-specific trend strengths k_1 and k_2 , giving a compact summary of the fitted gene-expression trajectory. The randomized residuals ask a different question: after the trend has been fitted, do the conditional count margins still look adequate on the uniform scale? In this demonstration, the Poisson trend captures an interpretable trajectory but leaves strong residual non-uniformity, while the negative-binomial trend greatly reduces both the one-sample residual discrepancy and the pseudotime-bin discrepancy. Across the 10 fitted demo genes, the NB margin has a smaller one-sample residual KS distance than the Poisson margin for all 10 genes; the median residual KS distance decreases from 0.278 to 0.035, the median quartile-indexed KS distance decreases by 0.112, and the median negative log-likelihood decreases by 1176.5. The comparison illustrates why trend interpretability and distributional adequacy should be assessed separately. These descriptive comparisons do not establish nominal test calibration for the scGTM fitting procedure.

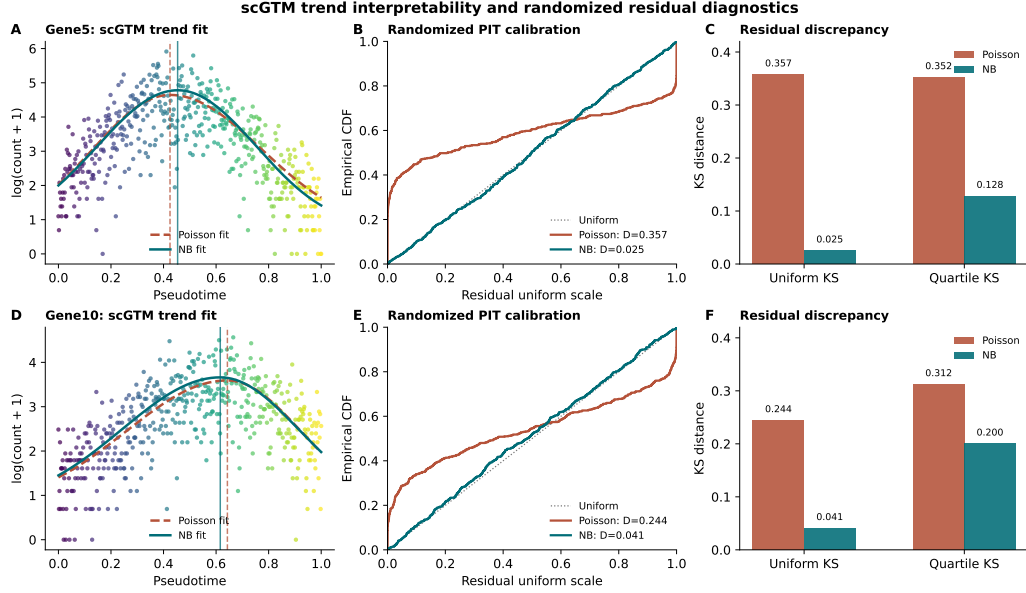
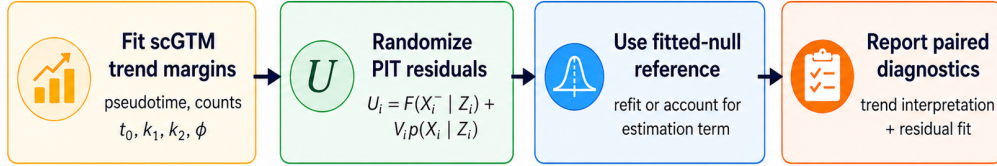


Figure 2: scGTM demonstration using two representative genes from a supplied 500-observation, 10-gene count matrix. Panels A and D show fitted Poisson and negative-binomial (NB) scGTM trends on the pseudotime scale, with vertical lines marking the estimated change time t_0 . Panels B and E show randomized PIT residual ECDFs against the uniform reference. Panels C and F compare the one-sample uniform KS distance with the largest two-sample KS distance between pseudotime quartiles. The uniform p -values reported in Table 6 are descriptive because the trend margins were fitted.



The residual diagnostic is descriptive unless the fitting step is included in the reference law.

Figure 3: Operational scGTM residual-diagnostic workflow. The trend parameters summarize the fitted pseudotime trajectory, while randomized PIT residuals diagnose the fitted conditional count margin. The residual statistic is descriptive unless the reference law also includes the same fitting step, for example through a refitted bootstrap.

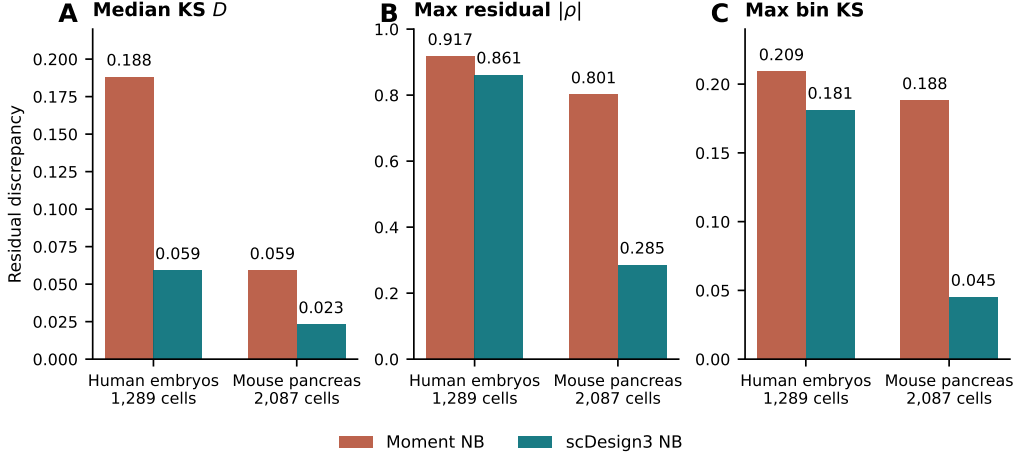


Figure 4: Residual diagnostics for human preimplantation embryos (1,289 cells, 10 genes; E-MTAB-3929) and mouse pancreatic endocrinogenesis (2,087 cells, 100 genes; GSE132188), using the subsets distributed with scDesign3. Moment negative-binomial margins are compared with scDesign3 conditional negative-binomial fits. (A) Median gene-wise uniform KS distance. (B) Maximum absolute pairwise residual correlation. (C) Maximum two-sample KS distance between pooled residuals in pseudotime quartiles. Panels B and C are descriptive summaries, not calibrated joint tests.

5.2. Real-Data Applications

Mouse pancreatic endocrinogenesis. The pancreatic data originate from a mouse pancreatic endocrinogenesis study and are publicly available as [GEO GSE132188](#). The processed trajectory data are distributed through scVelo and used by the scDesign3 tutorial ([Bastidas-Ponce et al., 2019](#); [Bergen et al., 2020](#); [Song et al., 2024](#)). The tutorial retains 2,087 cells and the top 100 highly variable genes with matched pseudotime. Figure 4 compares simple moment margins with scDesign3 conditional negative-binomial margins. The conditional fit reduces the median gene-wise KS distance from 0.059 to 0.023, the maximum absolute residual correlation from 0.801 to 0.285, and the maximum pseudotime-quartile KS distance from 0.188 to 0.045.

We then apply Algorithm 1 to the first 10 highly variable genes in the package ordering. For each gene, we fit the scDesign3 conditional negative-binomial margin, simulate 1,999 conditional count vectors at the observed pseudotimes, and repeat the fit using scDesign3’s per-gene regression function, `fit_marginal`. Fresh randomized PIT residuals are generated and the gene-wise KS statistic is recomputed after each fit. All 19,990 bootstrap fits succeeded. Table 7 compares the naive and refitted tail probabilities for the

same observed residuals. For *Chga*, the naive uniform p -value is 0.0237 and the refitted value is 0.0020. Both are below 0.05 before multiplicity correction, but only the refitted value remains below 0.05 after Holm correction over the 10-gene panel (0.0160, versus 0.1657 for the naive values). This difference concerns the reference distribution of the residual statistic, not an improvement in the observed fitted curve. Five genes, *Pyy*, *Iapp*, *Rbp4*, *Chga*, and *Nnat*, have Holm-adjusted refitted $p < 0.05$.

This analysis conditions on the supplied pseudotimes and gene ordering. Neither trajectory estimation nor highly variable gene selection is repeated in the bootstrap. It therefore assesses the selected conditional margins, without accounting for uncertainty in those upstream steps. The refitted bootstrap reference distributions are empirical approximations: the simulator-specific conditions for the asymptotic result have not been established here. Table 5 supplies a focused finite-sample check of the same spline-margin configuration under a specified fixed-design null, not a validation of the full pancreatic preprocessing workflow. Supplementary Appendix H reports conditional Monte Carlo intervals and the fixed randomization protocol. A single observed randomization is used per gene; sensitivity to that randomization is not assessed by increasing the number of bootstrap draws.

Table 7: Refitted marginal calibration for the first 10 highly variable genes in the 2,087-cell pancreatic endocrinogenesis data. Each refitted value uses $B = 1,999$ complete scDesign3 conditional-margin refits. Holm correction is over all 10 genes. The minimum attainable Monte Carlo p -value is 0.0005.

Gene	KS D	Naive p	Refit p	Holm p	Refit 95% cutoff
<i>Pyy</i>	0.0300	0.0472	0.0025	0.0160	0.0225
<i>Iapp</i>	0.0332	0.0203	0.0020	0.0160	0.0239
<i>Chgb</i>	0.0206	0.3395	0.1225	0.3675	0.0235
<i>Rbp4</i>	0.0439	0.0006	0.0005	0.0050	0.0232
<i>Spp1</i>	0.0241	0.1763	0.0715	0.3075	0.0250
<i>Chga</i>	0.0326	0.0237	0.0020	0.0160	0.0229
<i>Cck</i>	0.0167	0.6068	0.2885	0.3675	0.0217
<i>Ins1</i>	0.0224	0.2479	0.1520	0.3675	0.0263
<i>Nnat</i>	0.0432	0.0008	0.0005	0.0050	0.0233
<i>Ins2</i>	0.0263	0.1105	0.0615	0.3075	0.0272

Human preimplantation embryos. The human preimplantation embryo data were reported by [Petropoulos et al. \(2016\)](#) and are available under [E-MTAB-3929](#). We use the scDesign3 subset containing 1,289 cells and 10 genes. All supplied cell identifiers and developmental-stage labels match the public sample annotation. The five groups represent embryonic days 3–7, not five

inferred cell types; the supplied pseudotime ranges from 0 to 1. The conditional margin reduces the median gene-wise KS distance from 0.188 to 0.059, but the maximum absolute pairwise residual correlation remains 0.861. This contrast illustrates Proposition 3.9 and Remark 3.11: marginal uniformity, residual dependence summaries, and covariate-indexed residual structure are different diagnostic targets. Supplementary Appendix H specifies the package objects, provenance check, and numerical files in Online Resource 1.

Translational case study: IMvigort210 baseline PBMCs. We next analyze public integer UMI-count matrices from the baseline PBMC single-cell experiment reported by Yuen et al. (2020). The 10 patients, five responders and five nonresponders, were treated in the IMvigort210 atezolizumab program; the relevant cohort is registered as NCT02108652 and the single-cell matrices are deposited as GEO GSE145281. We independently retain cells with at least 500 UMIs, 200–6000 detected genes, and mitochondrial fraction at most 0.20. This leaves 24,134 of 26,749 candidate count-matrix columns. Response labels are used only to describe the samples, not to test treatment efficacy or construct a response classifier.

For a source-motivated panel fixed before residual fitting from the source study’s IL-8 and antigen-presentation analysis, we fit the transparent working margin

$$Y_{ijg} \sim \text{NB}(\mu_{ijg}, r_g), \quad \mu_{ijg} = L_{ij}\lambda_{jg},$$

where L_{ij} is the observed library-size offset for cell i in patient j , λ_{jg} is a patient-specific rate, and r_g is a shared gene-specific dispersion. We estimate each rate by its exposure-adjusted patient mean and r_g by the pooled NB2 moment equation; the same rule is repeated in every bootstrap sample. The diagnostic statistic

$$S_g = \max_{1 \leq j \leq 10} \sqrt{n_j} D_{jg}$$

uses the largest patient-stratified randomized-PIT KS discrepancy on its null fluctuation scale, avoiding dilution of a patient-specific discrepancy in a pooled empirical distribution. Each of $B = 9,999$ bootstrap replicates holds the patient and library-size design fixed, simulates counts, re-estimates λ_{jg} and r_g , generates fresh randomizers, and recomputes S_g . A fixed-parameter comparator omits only the re-estimation step.

To investigate the discrepancies under this shared-dispersion model, we also fit an exploratory extension with patient-specific dispersions r_{jg} . The mean model, count data, QC rules, and complete 10-gene panel are unchanged.

This extension allows patient-to-patient differences in conditional variance without introducing unverified cell-type annotations. Its moment estimators are refitted in each of a separate set of 9,999 bootstrap samples per gene. The extension and its simulation size were fixed before computation.

The working reference assumes independent cells conditional on patient and the retained library-size offsets. Patient stratification does not account for arbitrary within-patient dependence. Because library sizes and QC are derived from the expression data and then held fixed, this is a conditional working-model check, not a calibration result for the full preprocessing or patient-sampling procedure. A rejection can reflect a failure of these sampling assumptions as well as an inadequate count margin.

Figure 5 shows strong inadequacy of this simple working margin for *CXCL8*, *B2M*, *CD74*, *HLA-A*, *HLA-C*, *HLA-DQB1*, and *HLA-DRB1*: all seven refitted p -values equal the attainable minimum 0.0001 and equal 0.0010 after Holm correction over the 10-gene panel. Each has zero exceedances in 9,999 draws; the upper endpoint of its conditional 95% Monte Carlo interval for the unadjusted tail probability is 0.000369. The refitted values for *HLA-DQA1*, *TAP1*, and *TAP2* are 0.7785, 0.1188, and 0.1470, respectively.

Allowing patient-specific dispersion produces a mixed improvement. For *CXCL8*, S_g decreases from 8.768 to 5.403 and the refitted p -value increases to 0.0077; for *B2M*, the corresponding changes are 8.849 to 3.275 and $p = 0.0023$. Their mean fitted log scores also increase. However, the observed discrepancies increase for *CD74* and *HLA-C*, and the fitted log score decreases for *HLA-DQB1*. All seven originally flagged genes retain Holm-adjusted $p < 0.05$ under the extension (range 0.0010–0.0308). Neither candidate rejects the other three genes after adjustment. All 199,980 bootstrap refits across both models succeed. Patient-specific dispersion alone does not remove these discrepancies; Appendix G reports the complete comparison.

6. Applicability and Computational Considerations

The asymptotic result in Section 3 is most directly applicable to regular finite-dimensional fits. Modern single-cell tools differ substantially in whether their estimators and residual maps satisfy those conditions. Table 9 therefore separates settings in which the assumptions can plausibly be verified from settings in which a refitted simulation is only an empirical, workflow-specific approximation. These conditions determine the inferential interpretation of the diagnostic.

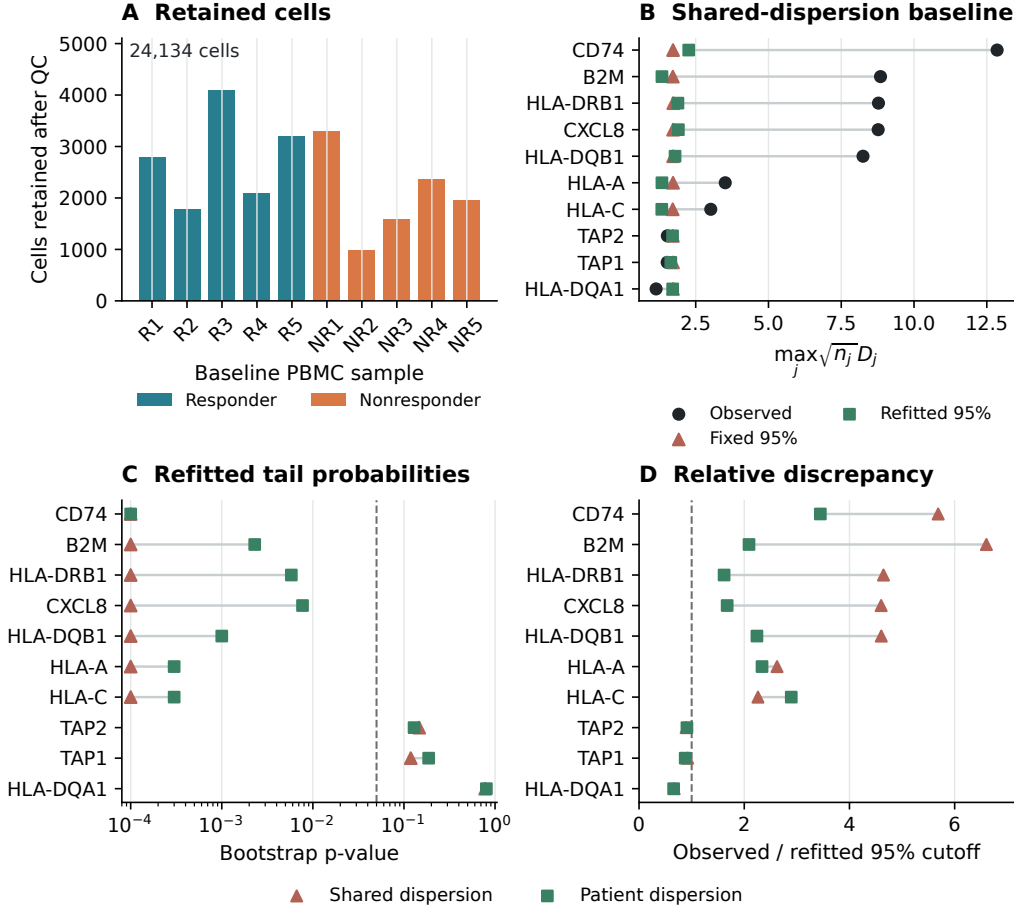


Figure 5: Public IMvigor210 baseline-PBMC residual diagnostics (GEO GSE145281; NCT02108652). Panel A shows cells retained after independent QC; response labels are descriptive only. Panel B compares the observed patient-stratified statistic S_g with 95% fixed-parameter and refitted-bootstrap cutoffs for the shared-dispersion model. Panel C compares plus-one refitted-bootstrap p -values under shared and patient-specific dispersions. Panel D compares the observed statistic divided by each model’s refitted 95% cutoff. Dashed lines in C and D mark unadjusted 5% thresholds, not multiplicity-adjusted decisions. Each model uses $B = 9,999$ replicates per gene. The gene panel was fixed before residual fitting from the source study’s CXCL8 and antigen-presentation focus. Supplementary Appendix G gives both models’ numerical results, Holm adjustments, and Monte Carlo intervals.

Let $C_{\text{fit}}(n, G)$ denote the cost of fitting the model to n cells and G genes, and let $C_S(n, G)$ be the cost of constructing residuals and the chosen statistic.

Table 8: Summary of diagnostic targets, numerical evidence, and supported interpretations.

Target	Key evidence	Supported interpretation and limitation
Fitted margins need fitted-null calibration	NB simulation: naive rejection 0.000; refitted rejection 0.040–0.062 across $n = 250$ –10,000	The fitted workflow, not the simple-null bridge, is the relevant reference; synthetic count margin.
Marginal KS misses copula errors	Wrong copula $\rho = 0.50$: marginal KS 0.054/0.050; residual-correlation diagnostic 0.663	Single-gene marginal residuals do not validate dependence; two-gene copula illustration.
scGTM trend fit and residual fit are distinct	10 demo genes: median residual KS 0.278 to 0.035; NB lower in 10/10 genes	Trend parameters and residual adequacy should be reported together; package demo matrix.
Pancreatic conditional margins require fitted calibration	<i>Chga</i> : naive $p = 0.0237$, refitted $p = 0.0020$; 19,990/19,990 bootstrap fits succeeded	The reference changes the tail probability and multiplicity-adjusted decision; not validation of the fitted copula.
Clinical PBMC margins remain falsifiable after calibration	IMvigor210: 7/10 genes have Holm $p = 0.0010$ under shared dispersion and $p \leq 0.0308$ with patient-specific dispersion	Allowing patient-specific dispersion does not remove the seven discrepancies; the model comparison is exploratory, not a treatment-response analysis.
Marginal improvement is not joint calibration	Human embryos: median KS 0.188 to 0.059, but residual correlation remains 0.861	Diagnostics must separate marginal, dependence, and covariate-indexed targets; 10-gene developmental subset.

A serial refitted bootstrap with B samples costs

$$O\{(B + 1)[C_{\text{fit}}(n, G) + C_S(n, G)] + BC_{\text{sim}}(n, G)\},$$

where C_{sim} denotes the cost of generating one bootstrap data set. Bootstrap replicates are conditionally independent, so they can run on separate workers. Residuals and gene-wise statistics can also be streamed so that all $n \times G$ residuals need not be stored simultaneously. Table 10 reports measured times for the observed procedure and one bootstrap replicate.

The conditional-regression study comprises $4 \times 500 = 2,000$ outer samples and $4 \times 500 \times 199 = 398,000$ bootstrap margin refits, or 400,000 fits including the observed samples. The timed four-scenario run took 3,122.6 seconds (52.0 minutes) on an Apple M1 Max CPU (10 cores), with 64 GiB memory and macOS 26.5.2, using four worker processes and one fitting core per worker. The largest conditional sample has 2,087 cells: the refit total describes computational workload, not validation at a 398,000-cell sample

Table 9: Relationship between representative single-cell workflows and the regularity conditions used by the randomized residual process. “Potentially verifiable” means that model- and algorithm-specific verification is still required.

Workflow	Relevant fitted object	Status of stated theory	Recommended interpretation
scGTM	Low-dimensional gene-wise trajectory and count-margin parameters	Potentially verifiable for fixed pseudotime, interior parameters, and a stable optimum; estimated pseudotime requires an additional expansion	Verify analytic or bootstrap conditions; otherwise assess the refitted reference empirically.
scDesign2	Gene-wise margins and an estimated Gaussian copula	Margin theory may be checked gene by gene; increasing-dimensional copula estimation is not covered by the fixed-dimensional statement	Calibrate marginal and dependence statistics separately and refit both components relevant to each statistic.
scDesign3	Smooth conditional margins, distribution selection, and dependence estimation	Smoothing, model selection, and increasing parameter dimension make asymptotic linearity and the plug-in remainder nonautomatic	Treat the full refitted workflow as empirical calibration unless sufficient conditions are established for the selected fit.
scVI-type models	Variational neural estimation with latent variables and many parameters	Not covered directly by the stated finite-dimensional regularity conditions	Use held-out predictive checks or a model-specific simulation study; describe a refitted bootstrap as heuristic unless separately justified.
Foundation models (e.g., scGPT)	Learned representations or predictions without an explicit fitted count CDF	A randomized PIT residual may not be defined	Outside the direct scope of this method until a probabilistic count reference and diagnostic target are specified.

size. Supplementary Appendix H.6 reports the software environment and task-level timings for these simulations, the 19,990 pancreatic refits, and the 199,980 clinical PBMC refits. Summed task times are distinguished from elapsed wall time.

Table 10 shows approximately linear observed runtime over the tested range, up to one million cells. Moment fitting is linear in n , whereas the implementation sorts residuals for KS evaluation, with $O(n \log n)$ comparison-sorting cost per gene. This is a feasibility check, not evidence that a full scDesign3 or variational model can be refitted at the same rate. For expensive models, wall time is dominated by C_{fit} ; parallel bootstrap workers can reduce elapsed time without reducing the number of fits; no parallel speedup ratio is

Table 10: Single-worker CPU scalability check for a lightweight 10-gene moment-NB workflow on the machine described in the text. Times are medians of five sequential repetitions, after an excluded warm-up. “Observed” includes generation of the synthetic counts, fitting, randomized residual construction, and the KS statistic; “one refit” repeats those operations on one bootstrap data set. The projected time for $B = 199$ is calculated from the two median timings and assumes serial execution; 199 complete refits were not timed. The memory column is an analytic baseline of $24n$ bytes for three 64-bit arrays, not measured peak memory; it excludes additional temporary arrays, interpreter overhead, and input-matrix storage. These timings do not represent the substantially more expensive scDesign3 fit.

Cells	Genes	Observed (s)	One refit (s)	Projected $B = 199$ (min)	Array baseline (MiB)
1,000	10	0.002	0.002	0.006	0.02
10,000	10	0.021	0.020	0.067	0.23
100,000	10	0.187	0.185	0.617	2.29
1,000,000	10	2.010	1.993	6.643	22.89

estimated here. Appendix H.6 specifies the timing boundaries and retains all five repetitions.

For models fitted by stochastic optimization, such as the scGTM likelihood applications in Cui et al. (2024), the initialization rule, parameter bounds, stopping criterion, and search budget are part of the fitting procedure. A refitted reference should use the same rules as the observed fit, including the treatment of optimizer randomization. Repeating these rules does not itself establish bootstrap consistency: the estimator must still satisfy the stability and asymptotic conditions stated above.

For smooth, low-dimensional models, the covariance expansion suggests a faster alternative to complete refitting. For randomized residuals, define

$$q(u) = \mathbf{1}\{T_{F_{\theta_0}}(X, V) \leq u\} - u, \quad a(u) = \dot{G}_{\theta_0}(u).$$

The fitted residual limit has covariance

$$K(u, v) = \mathbb{E}[\{q(u) + a(u)^T \psi_{\theta_0}(X)\} \{q(v) + a(v)^T \psi_{\theta_0}(X)\}].$$

Estimating K on a threshold grid and simulating centered Gaussian vectors gives an approximate reference for the maximum absolute residual process. This retains the covariance between the empirical and estimation terms; simply adding independent parameter noise would not. It avoids repeated fitting, but requires consistent derivatives, influence functions, and covariance estimates, together with control of grid approximation error. We describe this alternative analytically; no Gaussian-reference benchmark is reported in

this study. Sample splitting or cross-fitting can reduce reuse of the same cells, but changes the inferential target and does not by itself remove uncertainty in the fitted reference distribution.

7. Discussion

The central distinction is operational. If a discrete margin is known and the randomized distributional transform is used, the ordinary uniform KS reference is valid. If the margin is estimated from the cells being checked, the fitting step changes the residual process and must be represented in the reference law.

This distinction matters for fitted-model validation. A naive KS cutoff can be severely conservative under fitted models, as shown in Table 4. Under the regularity conditions described above, the remedy is to calibrate the statistic against the fitted workflow: simulate from the fitted model, refit the model in each simulated dataset, and recompute the same residual statistic. When the bootstrap analogues of the expansion hold, the refitted bootstrap targets the plug-in empirical-process limit rather than the simple-null bridge.

The pancreatic analysis shows why the distinction matters beyond null simulations. Refitting changes some gene-level p -values enough to alter the multiplicity-adjusted diagnostic decision. The simulation demonstrates severe conservatism for the tested moment-NB fit; it does not establish the direction of error for every fitted model. In the pancreatic example, some values also reach the Monte Carlo floor, which limits comparisons between very small tail probabilities.

The IMvigor210 analysis adds a clinically sourced stress test. Even after the reference procedure repeats the fitted patient-and-depth conditional margin, seven prespecified genes show discrepancies far beyond the refitted reference. The patient-specific-dispersion extension improves some fitted summaries but leaves all seven genes flagged after within-model Holm correction. Thus, relaxing shared dispersion alone is insufficient. The present analysis does not identify whether remaining discrepancies arise from cell-state mixtures, within-patient dependence, or other aspects of the count model. Its model comparison is exploratory and its log scores are in-sample summaries. This is evidence against the conditional working references, not evidence about atezolizumab efficacy or the published IL-8 biomarker claim. The distinction is important in translational analyses: reference calibration controls how a

diagnostic is judged, whereas biological interpretation requires a model and sampling design appropriate to the clinical question.

The contribution is broader than a package-specific diagnostic but narrower than a universal validation framework. For any workflow with an explicit fitted discrete or conditional distribution, the analyst can specify the residual map, identify whether its reference was fixed or fitted, and construct a reference for the same statistic under the same fitting algorithm. For latent-variable, foundation, or black-box generators, a probabilistic count reference and a refitting scheme must first be defined before KS-style residual calibration can be claimed.

The copula result adds a second warning. A gene-wise KS diagnostic is useful for checking marginal count calibration, but it cannot validate the dependence structure of a `scDesign2`-type simulator. Correlation and copula fit require joint residual diagnostics. For conditional single-cell trend-model and simulator diagnostics, the same logic suggests residual processes indexed jointly by residual thresholds and cell covariates such as pseudotime, modality, batch, or spatial region.

The supplementary appendix gives the fixed-null crossing formulas, an optional geometric interpretation, and the deferred proofs. For applications, the essential steps are those in Algorithm 1: specify the target, construct the residual statistic, and include the relevant estimation steps in its reference procedure. Report the statistic, calibration method, Monte Carlo resolution, and assumptions together. In pseudotime analyses, trend parameters describe the fitted trajectory, while residual diagnostics assess departures from the conditional count model. These outputs answer different questions and should be interpreted together.

Statements and Declarations

Declaration of generative AI and AI-assisted technologies in the writing process. During the preparation of this work the authors used OpenAI’s ChatGPT/Codex to assist with language editing, analysis-code drafting, LaTeX formatting, and consistency checks. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Competing interests. Yihao Li is affiliated with Genentech and Zhuang Liu is affiliated with Google. The IMvigor210 analysis uses only public GEO data

generated by the investigators of the cited study; no proprietary Genentech data or sponsor support was used. The authors declare no other competing interests directly related to this article.

Funding. No specific funding was reported for this work.

Author contributions. Elvis Han Cui led the study conception, methodology, implementation, manuscript drafting, and reproducibility packaging. Yihao Li and Zhuang Liu contributed to methodological framing, interpretation, and manuscript review and editing. All authors reviewed and approved the manuscript.

Ethics approval and consent to participate. Not applicable. This methodological article uses synthetic data and publicly available, previously published single-cell data.

Consent for publication. Not applicable.

Data and code availability. Online Resource 1 contains the editable manuscript sources, analysis scripts, numerical tables, saved bootstrap statistics, figures, and software versions. The README and Supplementary Appendix H give reproduction commands and implementation details. A public reproducibility snapshot, including the clean manuscript and supplement, is available at <https://github.com/ElvisCuiHan/randomized-ks-calibration>. The repository includes a per-file SHA-256 manifest identifying the distributed sources and numerical evidence.

The package examples use R’s `InsectSprays` data and the `scDesign3` objects `example_count`, `example_sce`, and `pseudotime`. These provide the mouse pancreatic subset (GSE132188; 2,087 cells, 100 genes) and human embryo subset (E-MTAB-3929; 1,289 cells, 10 genes). Embryo cell identifiers and developmental stages were verified against the public annotation, without reconstructing preprocessing or pseudotime estimation. The `scGTM` demonstration uses its distributed count matrix and trend-fitting function; Appendix H records the input path and source commit.

The clinical case study uses the 10 public baseline-PBMC integer count matrices in GEO GSE145281 from IMvigor210 (NCT02108652), retaining 24,134 cells after QC. Online Resource 1 includes source hashes, QC summaries, complete results for both dispersion models, and a script to download the source matrices. Clinical response labels are descriptive only. All 19,990

pancreatic and 199,980 PBMC refitted-bootstrap statistics are preserved, together with seeds and conditional Monte Carlo summaries.

References

- Babu, G.J., Rao, C.R., 2004. Goodness-of-fit tests when parameters are estimated. *Sankhyā: The Indian Journal of Statistics* , 63–74.
- Bastidas-Ponce, A., Tritschler, S., Dony, L., Scheibner, K., Tarquis-Medina, M., Salinno, C., Schirge, S., Burtscher, I., Böttcher, A., Theis, F.J., Lickert, H., Bakhti, M., 2019. Comprehensive single cell mrna profiling reveals a detailed roadmap for pancreatic endocrinogenesis. *Development* 146, dev173849. doi:[10.1242/dev.173849](https://doi.org/10.1242/dev.173849).
- Bergen, V., Lange, M., Peidli, S., Wolf, F.A., Theis, F.J., 2020. Generalizing RNA velocity to transient cell states through dynamical modeling. *Nature Biotechnology* 38, 1408–1414. doi:[10.1038/s41587-020-0591-3](https://doi.org/10.1038/s41587-020-0591-3).
- Cui, E.H., Song, D., Wong, W.K., Li, J.J., 2022. Single-cell generalized trend model (scGTM): a flexible and interpretable model of gene expression trend along cell pseudotime. *Bioinformatics* 38, 3927–3934. doi:[10.1093/bioinformatics/btac423](https://doi.org/10.1093/bioinformatics/btac423).
- Cui, E.H., Zhang, Z., Chen, C.J., Wong, W.K., 2024. Applications of nature-inspired metaheuristic algorithms for tackling optimization problems across disciplines. *Scientific Reports* 14, 9403. doi:[10.1038/s41598-024-56670-6](https://doi.org/10.1038/s41598-024-56670-6).
- Del Barrio, E., Deheuvels, P., Van De Geer, S., 2007. Lectures on empirical processes. European Mathematical Society Zurich.
- Dunn, P.K., Smyth, G.K., 1996. Randomized quantile residuals. *Journal of Computational and Graphical Statistics* 5, 236–244. doi:[10.1080/10618600.1996.10474708](https://doi.org/10.1080/10618600.1996.10474708).
- Durbin, J., 1973. Weak convergence of the sample distribution function when parameters are estimated. *The Annals of Statistics* , 279–290.
- Khmaladze, E.V., 1981. Martingale approach in the theory of goodness-of-fit tests. *Theory of Probability & Its Applications* 26, 240–257. doi:[10.1137/1126027](https://doi.org/10.1137/1126027).
- Li, W.V., Li, J.J., 2019. A statistical simulator scdesign for rational scrna-seq experimental design. *Bioinformatics* 35, i41–i50. doi:[10.1093/bioinformatics/btz321](https://doi.org/10.1093/bioinformatics/btz321).
- Lilliefors, H.W., 1967. On the kolmogorov-smirnov test for normality with mean and variance unknown. *Journal of the American Statistical Association* 62, 399–402. doi:[10.1080/01621459.1967.10482916](https://doi.org/10.1080/01621459.1967.10482916).

- Petropoulos, S., Edsgård, D., Reinius, B., Deng, Q., Panula, S.P., Codeluppi, S., Plaza Reyes, A., Linnarsson, S., Sandberg, R., Lanner, F., 2016. Single-cell RNA-Seq reveals lineage and X chromosome dynamics in human preimplantation embryos. *Cell* 165, 1012–1026. doi:[10.1016/j.cell.2016.03.023](https://doi.org/10.1016/j.cell.2016.03.023).
- Rosenberg, J.E., Hoffman-Censits, J., Powles, T., van der Heijden, M.S., Balar, A.V., Necchi, A., et al., 2016. Atezolizumab in patients with locally advanced and metastatic urothelial carcinoma who have progressed following treatment with platinum-based chemotherapy: a single-arm, multicentre, phase 2 trial. *The Lancet* 387, 1909–1920. doi:[10.1016/S0140-6736\(16\)00561-4](https://doi.org/10.1016/S0140-6736(16)00561-4).
- Rosenblatt, M., 1952. Remarks on a multivariate transformation. *The Annals of Mathematical Statistics* 23, 470–472. doi:[10.1214/aoms/1177729394](https://doi.org/10.1214/aoms/1177729394).
- Song, D., Wang, Q., Yan, G., Liu, T., Sun, T., Li, J.J., Li, W.V., 2024. scdesign3 generates realistic in silico data for multimodal single-cell and spatial omics. *Nature Biotechnology* 42, 247–252. doi:[10.1038/s41587-023-01772-1](https://doi.org/10.1038/s41587-023-01772-1).
- Sun, T., Song, D., Li, W.V., Li, J.J., 2021. scdesign2: a transparent simulator that generates high-fidelity single-cell gene expression count data with gene correlations captured. *Genome Biology* 22, 163. doi:[10.1186/s13059-021-02367-2](https://doi.org/10.1186/s13059-021-02367-2).
- Sun, T., Song, D., Li, W.V., Li, J.J., 2023. Author correction: scdesign2: a transparent simulator that generates high-fidelity single-cell gene expression count data with gene correlations captured. *Genome Biology* 24, 32. doi:[10.1186/s13059-023-02884-2](https://doi.org/10.1186/s13059-023-02884-2).
- Van der Vaart, A.W., 2000. Asymptotic statistics. volume 3. Cambridge university press.
- Yuen, K.C., Liu, L.F., Gupta, V., Madireddi, S., Keerthivasan, S., Li, C., Rishipathak, D., Williams, P., Kadel, E.E., Koeppen, H., et al., 2020. High systemic and tumor-associated IL-8 correlates with reduced clinical benefit of PD-L1 blockade. *Nature Medicine* 26, 693–698. doi:[10.1038/s41591-020-0860-1](https://doi.org/10.1038/s41591-020-0860-1).